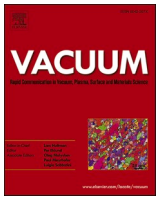
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Study of the optimal deposition conditions of Ag(O_2) and Ag(N_2) seed layers on ultrathin silver transparent conductive films

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ABSTRACT

In this study, ultrathin Ag layers were analyzed for their use as conductive films, which were fabricated via magnetron sputtering on a glass substrate and protected with aluminum-doped zinc oxide (AZO). These layers were obtained using seed layers made of silver deposited by adding small amounts of oxygen or nitrogen prior to the deposition of the metallic silver layer. This study aimed to determine the optimal amount of gas needed to maximize the conductivity and transparency properties of the conductive films with varied silver thicknesses. The optimal characteristics of the studied samples were verified using spectrophotometric, sheet resistance measurements, and surface morphology techniques. It was concluded that a smaller amount of silver required a higher amount of gas to be introduced into the seed layer. Besides, to fabricate an optimal conductive film, it is necessary to deposit 5.9 or 5.2 nm of silver with a seed layer obtained with a O_2 partial pressure of 1.7×10^{-4} mbar, or 5.2 or 4.7 nm of silver with a seed layer obtained using a N_2 partial pressure of 4.2×10^{-4} mbar. These samples provide sheet resistance values between 13.0 Ω/sq and 17.6 Ω/sq and transmission between 86.2 % and 87.3 % at 550 nm.

1. Introduction

Transparent conductive electrodes (TCEs) are widely used in applications such as solar cells [1,2], light-emitting diodes [3], touch screens [4], electrochromic devices [5], and transparent heaters [6]. Until recent years, the most widely used material was indium-doped tin oxide (ITO) because of its high visible transmittance (90 %) and low sheet resistance (10 Ω/sq), but due to the low abundance of indium on earth and other drawbacks such as its poor mechanical stability, new materials have been explored to improve the characteristics of TCEs. Alternatives to ITO that are widely studied include doped metal oxides, thin metals, conductive polymers, and nanomaterials (including carbon nanotubes, graphene, and metal nanowires) [7]. Specifically, thin metal films are an ideal material for TCEs because of their good photoelectric properties, simple deposition process via thermal deposition or sputtering, low cost, and excellent mechanical stability [8].

Silver is a material widely used for thin film deposition due to its abundance and simple deposition, as well as because it exhibits the highest electrical conductivity and the lowest optical loss over the visible spectral range compared to other metals. The lower the silver thickness, the greater the transmittance of the film; therefore, low silver thicknesses are necessary (less than 10 nm) to obtain TCEs. Silver is a material that, beyond a certain thickness known as the percolation threshold thickness, grows on substrates in the form of islands and in three dimensions that prevents good wettability over the entire surface, resulting in poor conductivity. This growth mode is consistent with the Volmer–Weber mode [9–11]. The initial stage of silver layer growth is characterized by the nucleation and formation of nano-islands or nanoclusters, which is accompanied by the coalescence phenomena of adjacent nano-islands. In the second stage, a percolated, non-continuous layer is formed. As the deposition process continues, the significance of percolation decreases until a continuous layer is established.

Silver conductivity increases with thickness. Therefore, to attain acceptable conductivities with a thin silver layer, it is necessary to apply a technique that enhances silver growth and achieves uniformity at low thicknesses, thereby reducing the percolation thickness. One such technique is the utilization of a material to dope the silver film and, thus, grow a silver-doped film with a smooth surface and good thermal stability [12–14]. An alternative strategy for controlling silver growth involves the selective introduction of alloying agents to influence specific stages of film development, for example Cu [15] or Au, Al, Cr, Mo, Ti [16]. Another technique is the introduction of a seed layer before the

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deposition of silver to attain a more homogeneous layer at low thicknesses. A wide variety of metals have been used as seed layers, including Al [17–19], Cr [20], Cu [21,22], Ge [23,24], Ni [25], Sn [26], Ti [27], and Zn [28]. Nevertheless, the disadvantage of using metals is that they are absorptive materials, leading to a drastic reduction in visible transmission. There is another technique to reduce the percolation thickness of ultrathin silver TCEs that is based on the addition of a gas during the growth of Ag to suppress the Volmer–Weber growth mode and, thus, improve the wetting ability of the silver on the substrate, including the use of oxygen [29,30] and nitrogen [31,32]. This technique can be used both for the growth of the entire silver thickness [29,31] and in the initial stages of growth as a seed layer for the subsequent silver metallic deposition [33].

In this study, we addressed the search for the optimal silver seed layer by introducing oxygen and nitrogen in small amounts during the initial silver deposition process in order to maximize the photoelectric properties of the entire ultrathin silver film. This approach allows minimal incorporation of oxygen or nitrogen in the seed layer, thereby enabling the homogeneous growth of the subsequent silver sputtered on it and, thus, reducing the threshold percolation thickness of the entire structure. In this way, we prepared ultrathin silver transparent conductive films through the sputtering process, and glass/Ag(O_2)/Ag/AZO or glass/Ag(N_2)/Ag/AZO structures were obtained (as a final layer, we deposited a protective layer of aluminum-doped zinc oxide (AZO) to optimize the transmittance). The optimal amounts of O_2 and N_2 were determined for each thickness of Ag. Additionally, a comparison was made between the results obtained using this technique of adding O_2 or N_2 during the growth of the silver seed layers, and those using Ge and GeO_2 seed layers.

2. Experiment

2.1. Preparation of ultrathin silver films

We fabricated all the samples in a semi-industrial (inline) high-vacuum system using magnetron sputtering (see Figure S1 a-b in the Supplementary Material). The system comprised three vacuum chambers and included 8 distinct magnetrons in the process chamber, with 4 on each side, thereby enabling the deposition of up to 4 different materials on each sample. The deposition system utilized DC pulsed power supplies with 600 mm × 100 mm rectangular targets. Mass flow controllers (MFCs) were used to control the amount of gas injected into the vacuum chamber. The system also included an ion gun to produce a cleaner substrate surface and improve the layer adhesion [34], and a cryogenic pump to reduce the residual water vapor. During the discharge, i.e., while the layer was being deposited, the substrate carrier moved through the process chamber at a constant speed, ensuring that each point on the substrate received identical exposure to the deposition process, thereby resulting in a uniform coating with a homogeneous thickness. Consequently, the thickness deposited on the substrate was inversely proportional to the carrier speed. The base pressure of the deposition system was 2×10^{-6} mbar in the process chamber before introducing the process gases, and the working pressure was between 2.0 and 2.8×10^{-3} mbar, depending on the material being deposited (in this deposition system, a flow of 200 standard cubic centimeters per minute (sccm) of Ar was approximately equivalent to a pressure of 1.8×10^{-3} mbar. In Ref. [35], the relationship between the pressure in the process chamber and the partial pressure of O_2 was revealed, as well as the stability in the total chamber pressure, by maintaining a constant total gas flow). The deposition was performed at room temperature. We used low-iron sodium calcium float glass (Pilkington Optiwhite™) substrates measuring 100 mm × 100 mm x 4 mm thick, which had been previously cleaned with a special glass detergent (ACEDET 5509).

Ultrathin silver films with AZO protection, glass/Ag/AZO, glass/Ag(O_2)/Ag/AZO, and glass/Ag(N_2)/Ag/AZO were deposited using the same high-vacuum deposition system. AZO was used to prevent Ag oxidation. Ag(O_2) and Ag(N_2) layers were sputtered by introducing a mixture of Ar and O_2 or N_2 gases during the sputtering process (with the proportion in flow between 6.3 % and 18.9 % for O_2 and 21–33.3 % for N_2. In Table S1 a-c of the Supplementary Material, we represent the measured working pressures for each gas flow introduced into the system and the measured partial pressures). Table 1 shows the deposition parameters for each material: condition 1 corresponds to Ag(O_2), condition 2 to Ag(N_2), condition 3 to pure Ag, and condition 4 to AZO.

These coatings, denoted as Ag(O_2) and Ag(N_2), refer exclusively to the formation of silver with the addition of O_2 or N_2 gas and not to the formation of oxidized or nitrided compounds. In the case of using oxygen during the growth of the seed layer, given that oxygen is a highly reactive gas, some oxygen atoms recombine with silver atoms, leading us to assume the formation of partially oxidized silver. However, for N_2 gas, being much less reactive, it remains unclear whether nitrogen atoms recombine with silver atoms and whether or not a compound is formed, as explained in Ref. [36].

Throughout the study, a protective layer of AZO was used to prevent the environmental oxidation of the ultrathin silver films. The samples produced to measure photoelectric properties were protected with 20 nm of AZO; therefore, the measurement of transmission and reflection spectra took this layer into account. For those samples whose surface morphology was analyzed, only 5 nm of AZO was used to minimize its influence; therefore, the morphology might differ slightly compared to the results presented in other authors' work.

Table 2 presents the O_2 flows that had been tested for each thickness of silver, and Table 3 presents the N_2 flows. For O_2, we started by testing very low flow rates, as silver has a high affinity for O_2, whereas for N_2, higher flow rates have been tested because N_2 is less reactive than O_2, requiring a larger amount of gas to obtain a compound that enhances the properties of ultrafine silver. To obtain Ag(O_2), deposition condition 1 shown in Table 1 was used, with a power of 110 W, an Ar flow of 300 sccm, and an O_2 flow between 20 and 70 sccm. For Ag(N_2), the parameters from condition 2 in Table 1 were used, with a power of 110 W, an Ar flow of 300 sccm, and a N_2 flow between 80 and 150 sccm. Metallic silver was deposited according to condition 3 in Table 1, with the power ranging from 110 to 400 W to achieve different thicknesses, and an Ar flow of 300 sccm. AZO was deposited with a power of 2000 W and an Ar flow of 300 sccm. Based on Tables 2 and 3, we obtained a total of 40 samples with Ag(O_2) and 36 with Ag(N_2).

All seed layers using O_2 were deposited with a thickness of 2.2 nm and those using N_2 with a thickness of 1.7 nm (these are the minimum thicknesses accepted by the deposition system). Thus, the total silver thickness in the structure is the sum of the seed layer's thickness and the metallic silver layer's thickness. For example, for a total thickness of 5.2 nm of Ag, a structure with an O_2 seed layer has a first layer of 2.2 nm of Ag(O_2) and a layer of 3 nm of pure Ag, and a structure with a N_2 seed layer has 1.7 nm of Ag(N_2) and 3.5 nm of pure Ag. Therefore, for lower thicknesses, the silver layer can only be obtained with an additive gas: the structures with thicknesses of 2.8 and 2.2 nm presented in Table 2 exclusively contain Ag(O_2), and the structures with thicknesses of 2.5, 2.0, and 1.7 nm presented in Table 3 exclusively contain Ag(N_2), without pure silver in these structures. Thus, for these thicknesses, there is no seed layer; instead, the entire silver layer was obtained with O_2 or N_2 additive gas.

The thicknesses of the metallic silver layers that were tested were those obtained when the sample holder traveled at the maximum speed allowed by the deposition system. This ensured that sputtering was performed in the shortest possible time to avoid impurities that might be present in the process chamber. Additionally, we chose the maximum discharge powers for sputtering because it is well known that high deposition rates achieve better growth of Ag layers [37]. By combining both parameters, carriage speed, and power, those thicknesses shown in Tables 2 and 3 were tested.

The O_2 flows selected for testing were those observed to improve the properties up to the point where no further improvement was obtained.

Table 1
Deposition parameters for each material.

No. Condition	Target Material	Target purity (%)	% weight	Power (W)	Power density (W/cm^2)	Ar Flow (sccm)	O_2 Flow (sccm)	N_2 Flow (sccm)
1	Ag	99.99	100	110	0.18	300	20–70	0
2	Ag	99.99	100	110	0.18	300	0	80–150
3	Ag	99.99	100	110–400	0.18–0.67	300	0	0
4	AZO	99.95	Zn 98 - Al 2	2000	3.33	300	0	0
5	Ge	99.99	100	100	0.17	200	0	0
6	Ge	99.99	100	500	0.83	80	120	0

Table 2
Ag thickness and O_2 flow for each tested sample.

Thickness of Ag (nm)	O_2 Flow (sccm)
8.2, 6.7, 5.9, 5.2	0, 20, 30, 40
4.5	0, 20, 30, 40, 50
3.9	0, 20, 30, 40, 50, 60, 70
2.8, 2.2	0, 30, 40, 50, 60, 70

Table 3
Ag thickness and N_2 flow for each tested sample.

Thickness of Ag (nm)	N_2 Flow (sccm)
6.7	0, 80, 100
5.2, 4.7	0, 80, 100, 120
4.0, 3.4, 2.5, 2.0, 1.7	0, 80, 100, 120, 150

The O_2 flows of 5 and 10 sccm were also tested at thicknesses between 5 and 8 nm, but they are not shown in the manuscript because they did not yield satisfactory results. Likewise, N_2 flows lower than 80 sccm were also tested at a thickness of 4.7 nm, but they are not presented in the manuscript as they did not result in improved layer properties. As can be observed, higher amounts of N_2 compared to O_2 are required as N_2 is less reactive than O_2.

To compare the good results obtained with these coatings, seed layers of Ge and GeO_2 were also tested, with the following structures and thicknesses:

glass/Ge(1 nm)/Ag(X nm)/AZO(20 nm) and glass/GeO_2(2 nm)/Ag(Y nm)/AZO(20 nm)

where X thickness of Ag is 6.0, 4.5, 3.0, and 1.7 nm, and Y thickness of Ag corresponds to 8.0, 6.0, 4.5, 3.0, and 1.7 nm.

The deposition parameters for Ge and the oxide of Ge are shown as conditions 5 and 6 in Table 1, respectively.

Hence, structures were prepared with different thicknesses of silver and different amounts of O_2 and N_2; in this way, we could find the gas flow that optimized the sheet resistance and visible transmission of the ultrathin silver films with different thicknesses of silver.

2.2. Characterization

Layer thicknesses were measured individually with a DektakXT® mechanical profilometer, which has a precision of approximately 1 nm. This measurement allows for the calculation of the exposure time required for the different thicknesses of each material, i.e., the speed of the deposition system carrier.

Optical measurements were performed with a UV–Vis/NIR spectrophotometer designed and built by the Photonic Technologies Group at the University of Zaragoza. The spectrophotometer can perform specular transmission and reflection measurements across the entire range of the solar spectrum from 300 to 2500 nm, with an angle of incidence of 8°.

Conductivity was measured via the 4-point probing method using a Jandel RM3000.

The performance of a transparent conductor is typically characterized by its figure of merit (FOM), which takes into account the two main properties of TCEs, namely, optical transmittance and sheet resistance. The goal is to achieve a balance between high transmittance and low resistance. Therefore, FOM allows samples to be compared with each other. Several figures of merit have been defined by researchers to quantify the performance of TCEs [38]. The most widely used FOM is the one defined by Haacke [39], which is shown in Equation (1). This is the FOM that was used in this study, and its formula is as follows:

$$FOM = \frac{T^{10}}{R_s} \tag{1}$$

where T is the transmittance, measured at 550 nm, and R_s is the sheet resistance (Ω/sq).

The surface morphology of the samples was assessed using Field-Emission Scanning Electron Microscopic (FE-SEM) images that were obtained using a Carl Zeiss MERLIN™. The surface morphology and roughness measurements were performed using Atomic Force Microscopy (AFM) Dimension Icon (Bruker).

3. Results and discussion

3.1. Samples with Ag(O_2) seed layer

Fig. 1a shows the sheet resistance (R_s) measured for all samples under study as a function of silver thickness (all data are shown in Table S2 a-b of the Supplementary Material). As can be seen in the figure, for thicknesses greater than 5.9 nm, the R_s is lower when no Ag (O_2) seed layer is used, while for the remaining thicknesses, its value decreases when a seed layer has been deposited beforehand. For the cases of 5.2 and 4.5 nm of Ag, the flow rate that results in the best R_s is 30 sccm; for 3.9 nm, it is 40 sccm; for 2.8 nm, it is 50 sccm; and for 2.2 nm, it is 60 sccm. From Fig. 1a, it can be concluded that the percolation threshold thickness decreases when a Ag(O_2) seed layer is used. Percolation threshold thickness is the thickness beyond which the film conducts electricity over macroscopic distances and hence sheet resistance exhibits a sharp drop from infinite value. We can only provide an estimate of the percolation thickness due to the limited data points on the curve and the minimum thickness achievable by our deposition system. For films of Ag without seed layer, this thickness can be estimate at 4.5 nm; for 20 sccm of O_2 at 3.9 nm; and for the remaining fluxes, this thickness falls below 2.2 nm, which is the minimum thickness permitted by the deposition system.

Fig. 1b illustrates the transmission measured at 550 nm for each O_2 flow as a function of silver thickness. The improvement in transmission when using a Ag(O_2) seed layer is evident. The best transmission was obtained when 30 sccm of O_2 was used for thicknesses greater than 4.5 nm; for the thickness of 3.9 nm, the best transmission was obtained at 40 sccm of O_2; and for thicknesses of 2.8 and 2.2 nm, it was obtained at 60 sccm. From Fig. 1a and 1b, it can be concluded that increasing the O_2 flow rate beyond 60 sccm does not improve the properties of the samples.

Fig. 2a shows the figure of merit (FOM) represented as a function of O_2 flow for different thicknesses of Ag. The FOM data for each gas flow were also plotted as a function of silver thickness (Fig. 2b).

As can be seen in both subfigures, a lower Ag thickness requires a higher amount of O_2 to achieve a better FOM, and each thickness

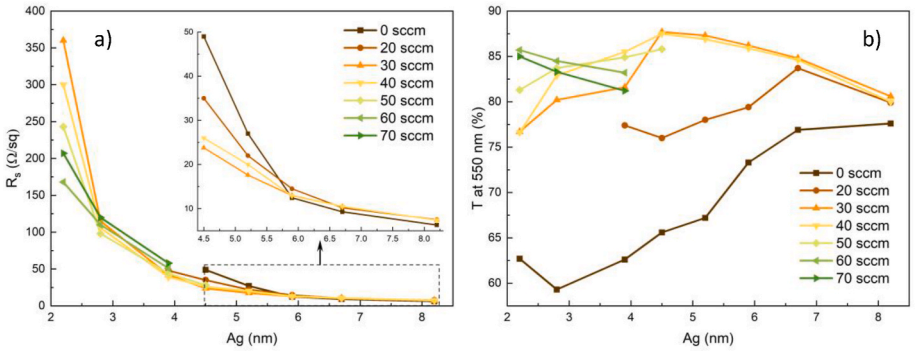


Fig. 1. (a) R_s and (b) T at 550 nm for each O_2 flow as a function of Ag thickness.

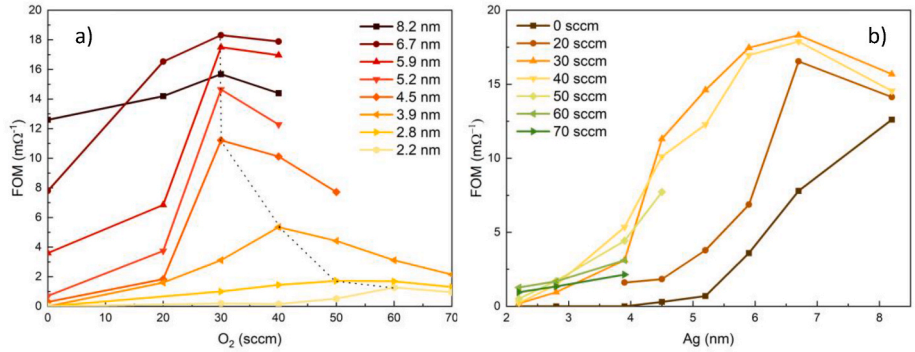


Fig. 2. (a) FOM for different Ag thicknesses as a function of O_2 flow; (b) FOM for different O_2 flows as a function of Ag thickness.

reaches an O_2 threshold beyond which no further improvement is observed. For silver thicknesses of 8.2, 6.7, 5.9, 5.2, and 4.5 nm, the optimal amount of O_2 is 30 sccm; for a thickness of 3.9 nm of silver, this amount is 40 sccm; for 2.8 nm, the sample with the best FOM was obtained at 50 sccm of O_2; and finally, for 2.2 nm, the optimal amount is 60 sccm. Of all the samples, the one with the best FOM is the sample with 6.7 nm of Ag and a seed layer deposited with 30 sccm of O_2.

As shown in Fig. 2 and according to the method used to calculate the FOM, an improvement in the FOM values was observed for all silver thicknesses when a Ag(O_2) seed layer was introduced. However, if we only examined the sheet resistance (R_s) and visible transmission (T_v) data, we could see that, for thicknesses of 8.2 and 6.7 nm, introducing oxygen during layer growth improved visible transmission but not sheet resistance. Therefore, for these thicknesses, it would be necessary to decide which of the two parameters is more important to maximize and, accordingly, choose whether or not to use a Ag(O_2) seed layer. In contrast, for the thickness of 5.9 nm, the minimal loss in R_s was compensated by the significant improvement in T_v obtained with the use of a seed layer with 30 sccm of O_2. Instead, both magnitudes improved with the use of a Ag(O_2) seed layer from 5.2 to 2.2 nm of silver.

As an example of one of the studied Ag thicknesses, the transmission spectrum measured at different O_2 flows for 3.9 nm of Ag is presented in Fig. 3a, and the transmission measured at 550 nm versus the sheet resistance is shown in Fig. 3b. The improvement in the properties of the entire structure by introducing exactly 40 sccm of O_2 is clearly evident. In the Supplementary Material, the transmission spectra measured for all the samples under study are presented in Fig. S2. In all the samples with a silver thickness of less than 6 nm and without a seed layer, an extraordinary absorption was observed between approximately 500 nm and 1500 nm. This phenomenon is attributed to silver being deposited in the form of nano-islands, producing plasmonic hot spots [40–42]. One of the main indicators that silver is deposited in a continuous ultrathin film is the disappearance of this absorption band.

It should be taken into consideration that, in this study, the reported transmission was measured including the contribution of the spectrum of the glass substrate. Therefore, in the calculation of the transmission of

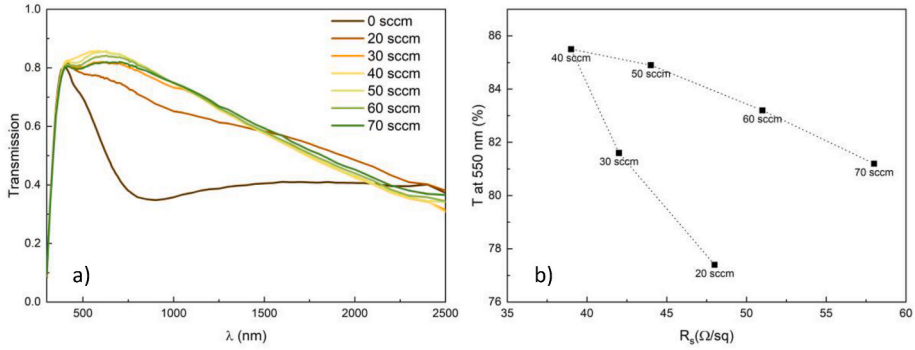


Fig. 3. (a) Transmission spectrum and (b) T at 550 nm versus R_s at different O_2 flows for 3.9 nm of Ag.

the thin silver layer, the reflection from the glass side, which is on the order of 4 %, was also considered (in the Supplementary Material, the transmission, reflection, and absorption spectra of the used glass are shown in Fig. S3). Hence, if the reflection spectrum of the glass was excluded from the calculation, the transmission value increased and reached 90 % for a silver thickness of 3.9 nm, which is consistent with the values reported in other studies such as [33].

The surface morphologies of the Ag and Ag(O_2)/Ag films with different thicknesses are shown in Fig. 4. Samples with a Ag(O_2) layer were selected with the optimal O_2 flow at each thickness, and they were protected with 5 nm of AZO, which was necessary to protect them from oxidation while minimizing the influence of the AZO layer on the surface morphology. As presented in Ref. [43], the obtained FE-SEM images reveal that thinner pure silver films exhibit a granular structure due to the Volmer–Weber growth of Ag in the form of islands. As thickness increases, the nanoclusters gradually become larger, transitioning from an island film to a layer film, and ultimately forming a continuous silver film. However, upon introducing a small amount of O_2 into the metal formation process, according to Ref. [33], the surface morphology changes to consist of irregular-shaped nanoclusters formed by the coalescence of regular and polygonal metal nanoclusters. Hence, it is necessary to introduce a seed layer to reduce the threshold thickness of the silver film, thereby enabling higher transmittance and better sheet resistance at a lower thickness, which fulfills the requirements of transparent conductive films.

Fig. S4 of the Supplementary Material shows the FE-SEM cross-sectional image of the 4.5 nm Ag sample without a seed layer and protected with 20 nm of AZO. This image reveals the granular structure exhibited by ultrathin Ag layers.

Fig. 5 shows the surface morphology of the 4.5 nm Ag samples with or without a seed layer obtained using AFM. These samples were protected with 5 nm of AZO. As show in the figure, in the case without a seed layer, the sample exhibits higher roughness compared to the sample with a Ag(O_2) seed layer; the lowest roughness was observed in the sample obtained with the optimal O_2 flow (Fig. 5c), which results in better photoelectric properties. The sample without a seed layer (Fig. 5a) exhibits larger and taller clusters, while incorporating the additive gas in the seed layer (Fig. 5b, 5.c, and 5.d) leads to the subsequent deposition of the silver layer producing clusters with a smaller volume and height. The sample generated with a lower amount of O_2 (Fig. 5b) produces a surface more similar to the sample without a seed layer (Fig. 5a), whereas if the optimal amount of O_2 is exceeded (Fig. 5d), the roughness and peak height increase.

As shown in Fig. 6, a similar behavior was observed for the samples produced with 2.5 nm of Ag and protected with 5 nm of AZO. A smoother surface was observed for the sample produced with O_2 additive (Fig. 6b), with clusters of smaller volume and height.

Table 4 summarizes the roughness properties of the six analyzed samples.

These results are consistent with those presented by Zhao et al. [33]. Their study reported an improvement in the TCE characteristics when using a partially oxidized silver seed layer, which allows the subsequent pure silver layer to be deposited into a completely continuous layer on a ZnO substrate. By contrast, our work focused on studying the optimal deposition conditions for Ag(O_2) seed layers deposited directly on glass rather than on ZnO.

According to Bauer [44,45], to achieve homogeneous growth of a metal instead of island growth, which occurs in the early stages, the formula $\gamma_m + \gamma_i < \gamma_s$ must be satisfied, where γ_m is the surface free energy of the metal, γ_i is the interfacial free energy between the metal and the substrate, and γ_s is the surface free energy of the substrate. As noted by Bauer [44], unless there is extraordinarily strong adhesion at the metal–substrate interface (which does not occur for noble metals such as silver), island growth is thermodynamically favorable for a high-surface-energy metal deposited on a low-surface-energy substrate. Thus, to improve metal growth, it is necessary to reduce either the surface free energy of the metal or the interfacial free energy between the metal and the substrate. According to Yun [8], incorporating a gas, such as oxygen in our study, in the initial stages of silver growth reduces the metal's surface free energy γ_m, leading to a decrease in the difference between $\gamma_m + \gamma_i$ and γ_s. The surface free energy is reduced by adding a gas because the material grows by forming irregular-shaped nanoclusters with smaller surface areas compared to those formed when only the metal grows [46], as can be seen in the FE-SEM images shown in Fig. 4. These irregular and elongated nanoclusters are formed as a result of a decrease in the rate of island coalescence [47,48]. In addition to the improvement in the metal's free energy upon adding an additive gas, there is also an enhancement in Ag–substrate adhesion, resulting in a reduction in interfacial free energy and thereby improving the subsequent deposition of metallic silver [8].

Therefore, the inclusion of an additive gas in the gas atmosphere inhibits coalescence-driven cluster reshaping, thereby facilitating in-plane island formation and promoting a two-dimensional growth morphology [32]. The improvement in metal growth is also attributable to the weakening of the driving force for cluster coalescence in the liquid-like clustering stage, as the suppression of the surface diffusion of metal atoms is driven by the reduced surface free energy [8].

This improved growth of silver on the Ag(O_2) seed layer results in a lower R_s value because a continuous layer is formed, providing sufficient electrical paths, and a higher T_v value since the lower number of nanoclusters with a smaller volume result in less scattering of the incident light on the sample surface. However, to achieve this goal, a minimum amount of oxygen is needed to allow the formation of irregular and in-plane nanoclusters with lower free energy, thus requiring a

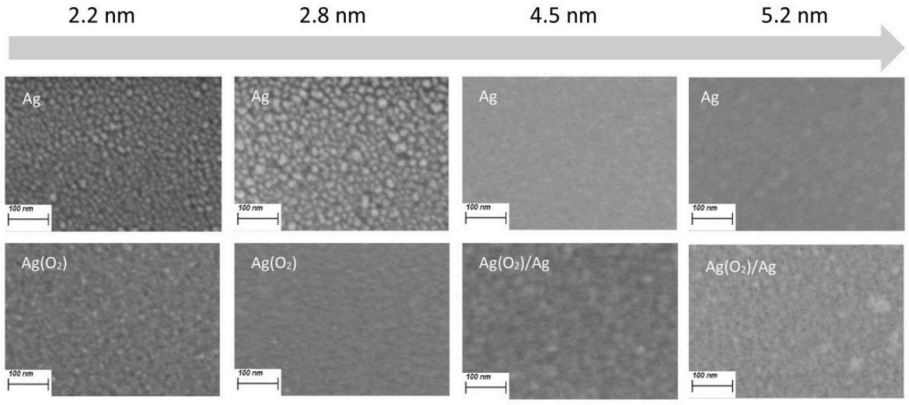


Fig. 4. Top-view FE-SEM images of the samples with different Ag thicknesses and with or without O_2 additive in the seed layer or in the Ag layer itself. The images were obtained using a 5 kV electron beam and an in-lens secondary electron detector.

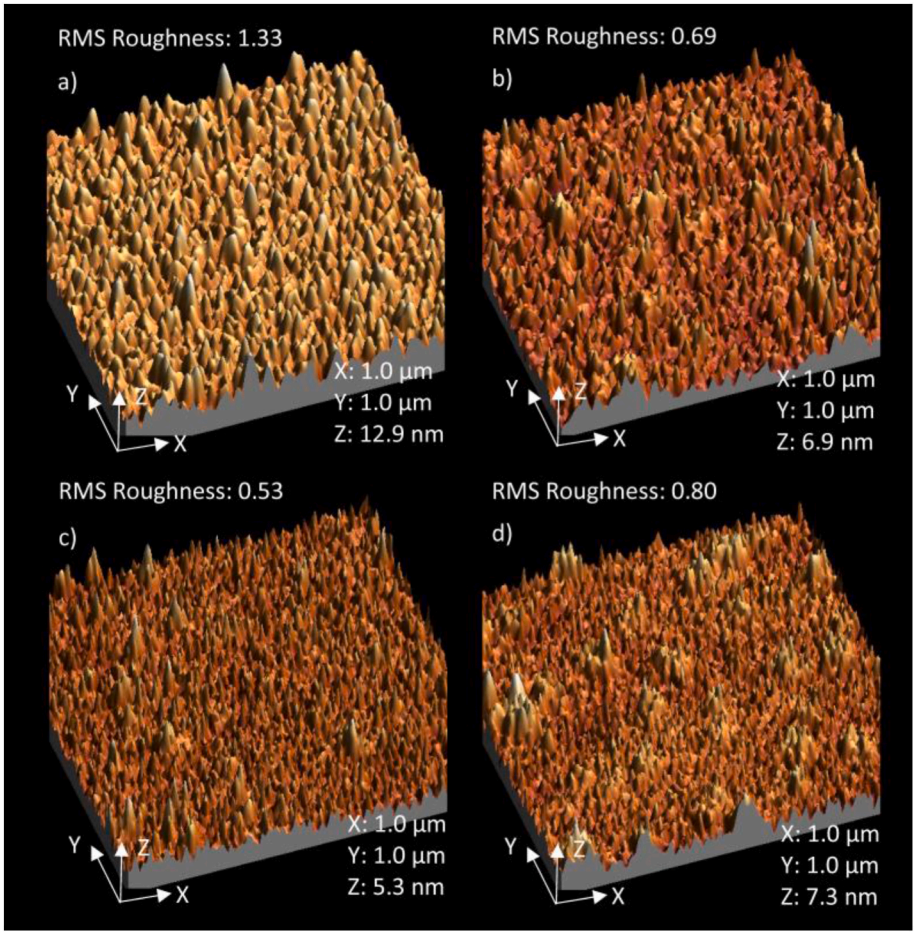


Fig. 5. AFM topography of (a) 4.5 nm Ag, (b) 4.5 nm Ag(O_2)/Ag with 20 sccm of O_2, (c) 4.5 nm Ag(O_2)/Ag with 30 sccm of O_2 (optimal flow), and (d) 4.5 nm Ag (O_2)/Ag with 50 sccm of O_2.

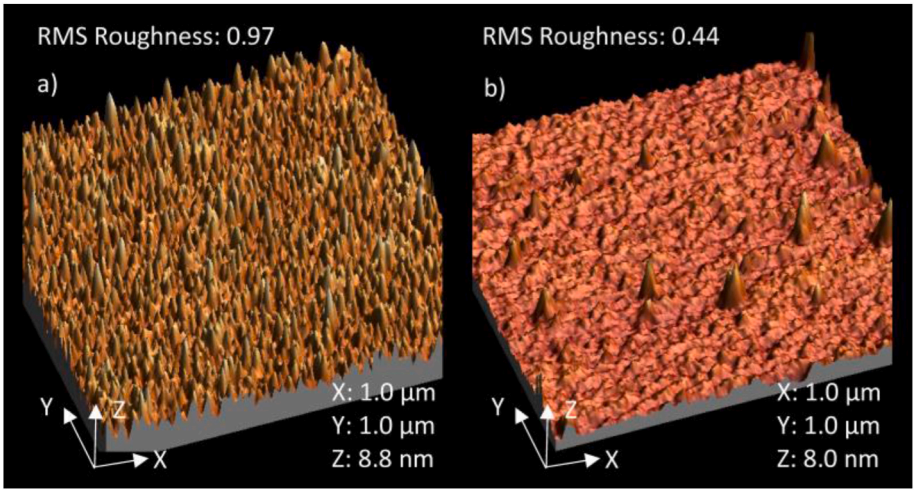


Fig. 6. AFM topography of (a) 2.5 nm Ag and (b) 2.5 nm Ag(O_2) with 50 sccm of O_2 (optimal flow).

higher amount of O_2 at lower Ag thicknesses, since there is a lower quantity of Ag atoms present in the atmosphere. If this optimal amount of O_2 is further increased, although there is still improvement in the photoelectric properties compared to the deposition of pure Ag, there is no improvement compared to when the optimal amount of O_2 is used because the insulating properties of silver oxide begin to appear, as there are many O_2 atoms available for Ag atoms to combine with and form silver oxide. This oxide formation should also result in an improvement in the visible transmission of the structure. However, this did not occur in our study, which is consistent with [29], although in our study there is a minimal reduction in T_v of no more than 1 %. This reduction is due to the increase in the roughness of the sample, as shown in Fig. 5d.

3.2. Samples with $Ag(N_2)$ seed layer

Fig. 7a shows the sheet resistance (R_s) for each tested N_2 flow in the seed layer as a function of silver thickness (all data are shown in Table S3 a-b of the Supplementary Material). As shown in the figure, for a thickness of 6.7 nm, a similar R_s was obtained with or without a seed layer, and for thicknesses lower than 5.2 nm, the improvement in R_s is notable with a seed layer. For thicknesses of 5.2 and 4.7 nm, the best R_s was achieved when 80 sccm of N_2 was used, and for thicknesses of 3.4 and 1.7 nm, it was achieved with 100 sccm of N_2. In contrast, for thicknesses of 4.0, 2.5, and 2.0 nm, the R_s values are similar when using 100 or 120 sccm of N_2. From this figure, we can estimate that the percolation thickness is 4.7 nm with pure Ag in the sample, whereas when a gas additive of N_2 is used during silver growth, this thickness is lower than the minimum thickness allowed by the system, which is 1.7 nm.

Fig. 7b illustrates the transmission measured at 550 nm for each N_2 flow as a function of silver thickness. The improvement in transmission when using a $Ag(N_2)$ seed layer is evident. For thicknesses of 6.7 and 5.2 nm, the highest transmission was obtained with 80 sccm of N_2, and for lower thicknesses, the optimal amount of N_2 is 100 sccm.

Fig. 8a and b presents the FOM data as a function of N2 flow for different silver thicknesses and the FOM data as a function of silver thickness for different flows of gas, respectively.

Like in the case of $Ag(O_2)$ seed layer, when using a $Ag(N_2)$ seed layer, a lower silver thickness requires a higher N_2 flow, but when the limit is reached, no further improvement is observed. Since nitrogen is a less reactive gas, in order to improve the properties of the structure, it is necessary to have a higher flow compared to the use of O_2. The sample with the best FOM is the one with 5.2 nm of Ag and a seed layer deposited with 80 sccm of N_2.

As an example, the transmission spectrum obtained at 4.0 nm of Ag for different N_2 flows is presented in Fig. 9a, and the transmission at 550 nm as a function of sheet resistance is presented in Fig. 9b. Both figures show that, for this thickness, the flow that provides the best properties is 100 sccm of N_2. In the Supplementary Material, the transmission spectra measured for all the samples under study are presented in Fig. S5.

Fig. 10 shows the surface morphology (AFM images) of the pure Ag samples with thicknesses of 2.5 nm and 4.5 nm compared to the 2.5 nm $Ag(N_2)$ and 4.5 nm $Ag(N_2)$/Ag samples. As is the case of the sample with a $Ag(O_2)$ seed layer and in accordance with [33], for both thicknesses, when a gas additive is used, the images should show a morphology consisting of irregular-shaped nanoclusters formed as a result of a decrease in the island coalescence rate [47,48], which translates into an improvement in the samples' photoelectric properties.

We can see the surface topography of the two samples obtained by adding N_2 during silver growth via AFM: Fig. 11a shows the sample with 4.5 nm of Ag, including 1.7 nm of $Ag(N_2)$ seed layer, and Fig. 11b shows the sample with 2.5 nm of $Ag(N_2)$. The roughness of both samples is lower than that of pure silver (Fig. 5a and 6a), and in both cases, clusters of smaller volume and height can be observed. However, compared to the optimal samples with a $Ag(O_2)$ seed layer (Fig. 5b and 6b), those with a $Ag(N_2)$ seed layer exhibit higher roughness.

Table 5 summarizes the roughness properties of the two analyzed samples.

Comparing the roughness data presented in Table 4 with those in Table 5, for the same thicknesses, we see that the samples with a $Ag(O_2)$ seed layer exhibit lower roughness than those with a $Ag(N_2)$ seed layer. The sample with 4.5 nm of Ag has a roughness of 0.53 when using a Ag (O_2) seed layer and 0.63 when the seed layer is $Ag(N_2)$. For the sample with 2.5 nm of Ag, the roughness is 0.44 with $Ag(O_2)$ and 0.51 with Ag (N_2). Therefore, the samples with $Ag(O_2)$ as the seed layer should have better photoelectric properties for the thicknesses analyzed by AFM.

In summary, just like a $Ag(O_2)$ seed layer, a $Ag(N_2)$ layer improves the photoelectric properties of ultrathin silver structures. In this case, even for 6.7 nm of Ag, both R_s and T_v are enhanced, which are contrary to what happened with $Ag(O_2)$. The same reasoning applied to the addition of O_2 during the growth of the $Ag(O_2)$ seed layer can be applied to $Ag(N_2)$; that is, there is an improvement in the metal's free energy, resulting in a homogeneous and smooth growth of silver, which consequently enhances the properties of the entire structure. These results are consistent with those presented by Ko et al. [31].

3.3. Comparison of samples with $Ag(O_2)$, $Ag(N_2)$, Ge, and GeO_2 seed layers

Samples with different Ag thicknesses were deposited using Ge and GeO_2 seed layers to compare the obtained results with the optimal Ag (O_2) and $Ag(N_2)$ samples. These samples were also protected with 20 nm of AZO, thus achieving an accurate comparison in transmission.

From the samples with $Ag(O_2)$ and $Ag(N_2)$ seed layers presented in the previous sections, the sample that provides the best R_s and transmission values was selected for each silver thickness. Fig. 12 shows the sheet resistance and transmission at 550 nm of the samples obtained with $Ag(O_2)$, $Ag(N_2)$, GeO_2, and Ge seed layers at different silver thicknesses (all data are shown in Table S4 a-b of the Supplementary Material).

We can see from these figures that transmission with the $Ag(O_2)$ seed layer is higher for all thicknesses, as this is the least absorbent material among the four studied materials. Regarding R_s, however, the $Ag(N_2)$ seed layer provides better values than the $Ag(O_2)$ seed layer, so the choice between these materials would depend on whether the application of these structures requires maximizing T_v or R_s. Concerning the GeO_2 seed layer, it provides R_s values similar to those provided by Ag (N_2), but the transmission values of $Ag(N_2)$ are higher than those of GeO_2 (except for 6 nm of Ag), indicating that $Ag(N_2)$ always offers better photoelectric properties. For the Ge seed layer, it provides worse values than the other materials owing to the visible absorbance exhibited by this material.

Fig. 13a shows the figure of merit for each seed layer as a function of

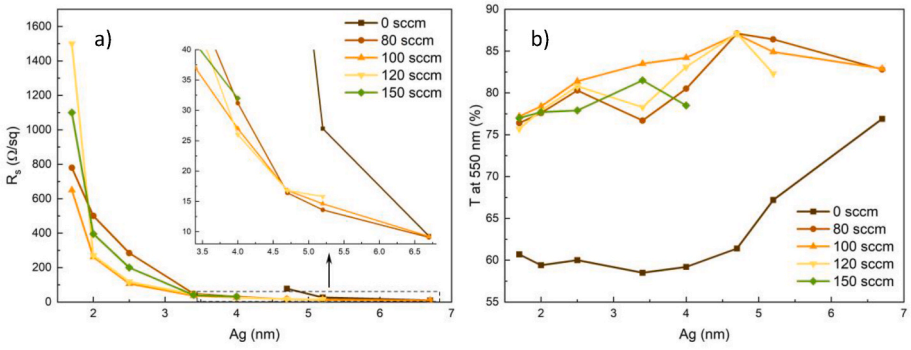


Fig. 7. (a) R_s and (b) T at 550 nm for different N_2 flows as a function of Ag thickness.

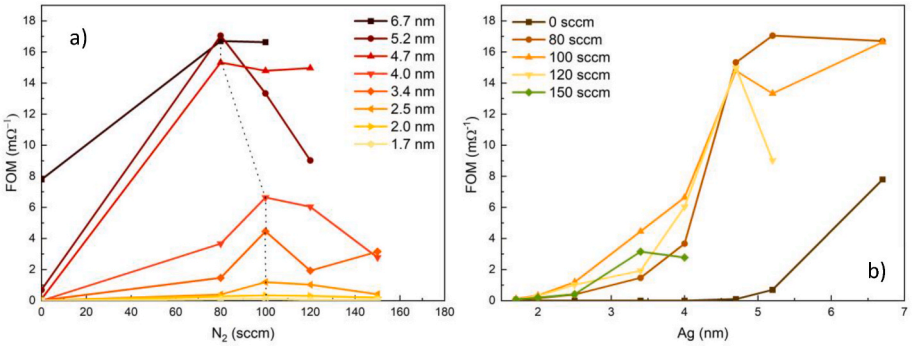


Fig. 8. (a) FOM for different Ag thicknesses as a function of N_2 flow; (b) FOM for different N_2 flows as a function of Ag thickness.

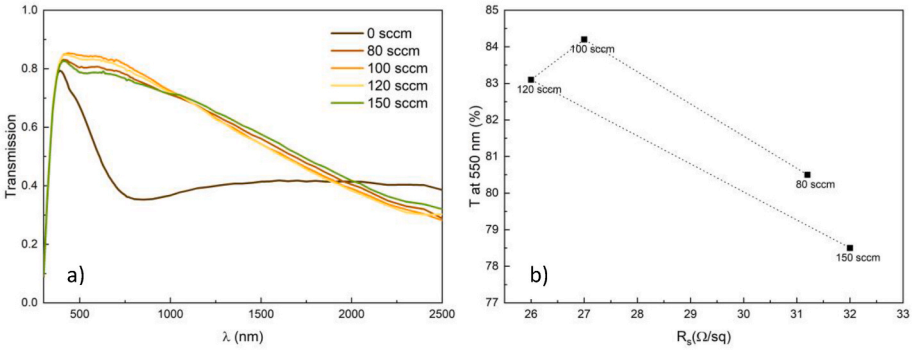


Fig. 9. (a) Transmission spectrum and (b) T at 550 nm versus R_s at different O_2 flows for 4.0 nm of Ag.

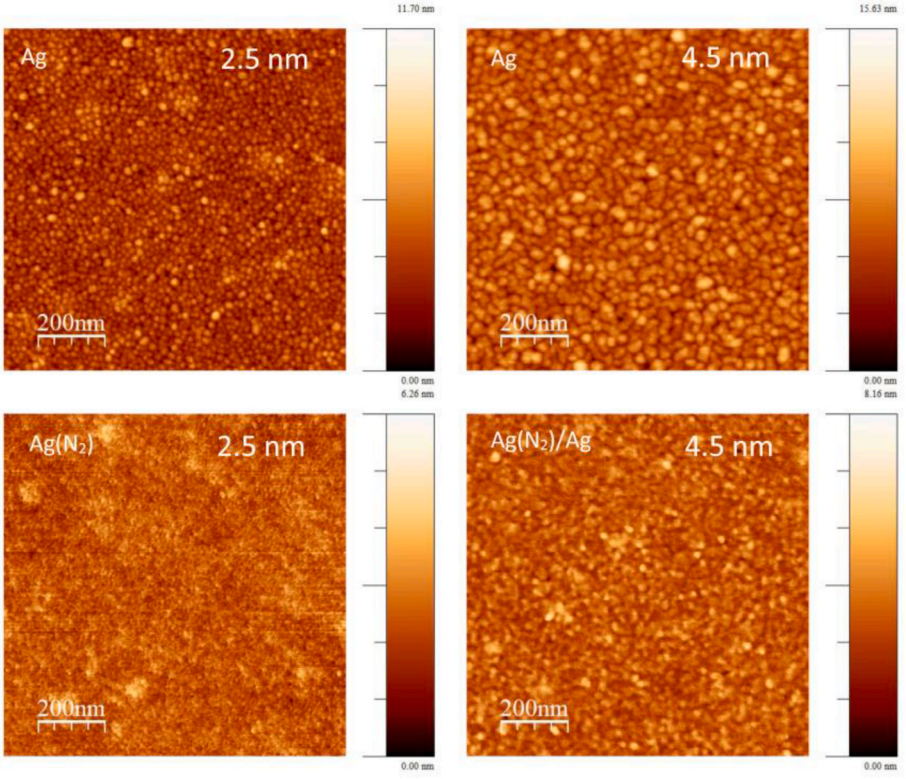


Fig. 10. Top-view AFM images of samples with different Ag thicknesses and with or without N_2 additive in the seed layer or in the Ag layer itself.

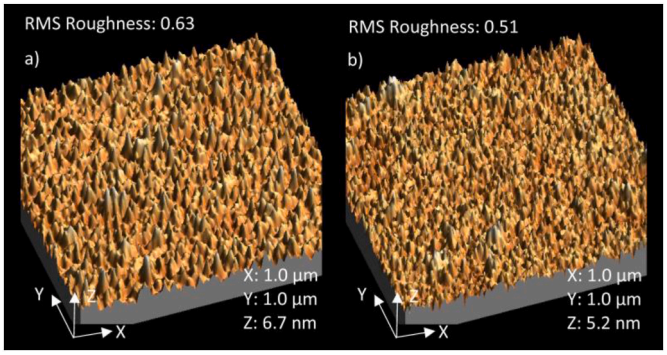


Fig. 11. AFM topography of (a) 4.5 nm $Ag(N_2)$/Ag with 80 sccm of N_2 and (b) 2.5 nm $Ag(N_2)$ with 100 sccm of N_2.

silver thickness. It can be seen that, between 2.5 and 5.2 nm, the highest FOM values are provided by the $Ag(N_2)$ seed layer; for thicknesses of 2.2, 6.7, and 8.2 nm, the highest FOM values are provided by the $Ag(O_2)$ seed layer; and for a thickness of 6 nm, the highest values are provided by the GeO_2 seed layer.

We obtained Fig. 13b by directly plotting the transmission at 550 nm versus R_s for each seed layer. For these types of structures, it is desirable to maximize T_v while minimizing R_s, so the goal is to be in the upper left part of the curve. From the figure, we deduce that the seed layers providing the best photoelectric properties are $Ag(O_2)$ and $Ag(N_2)$. Regarding thicknesses, for $Ag(O_2)$, the optimal thicknesses are 5.9 and 5.2 nm, and for $Ag(N_2)$, the optimal thicknesses are 5.2 and 4.7 nm. Again, the decision regarding thickness would depend on whether the application requires a higher T_v value or a lower R_s value, as there are two pairs of thicknesses (5.9 nm for $Ag(O_2)$ and 5.2 nm for $Ag(N_2)$, and 5.2 nm for $Ag(O_2)$ and 4.7 nm for $Ag(O_2)$) that provide good values of T_v and R_s. The first pair offers lower R_s values (13.0 Ω/sq and 13.6 Ω/sq) but also lower transmission (86.2 % and 86.4 %), while the second pair offers higher T_v values (87.3 % and 87.1 %) but also higher R_s values (17.6 Ω/sq and 16.4 Ω/sq).

In the Supplementary Material, the transmission spectra measured for all the samples under study with GeO_2 and Ge seed layers are presented in Fig. S6.

Therefore, the superior results obtained with $Ag(O_2)$ and $Ag(N_2)$ seed layers are evident in comparison to metallic seed layers, such as Ge. In the case of a Ge oxide seed layer, a better FOM is obtained only for a silver thickness of 6 nm compared to the other seed layers. At this thickness with this seed material, although there is no significant reduction in T_v compared to using $Ag(O_2)$ (only 1 %), the reduction in R_s is quite notable (from 13 Ω/sq with $Ag(O_2)$ to 11 Ω/sq with GeO_2), which makes the FOM value superior for the GeO_2 seed layer.

This study was conducted using a 4 mm thick glass substrate. If an anti-reflective coating were applied to the opposite side of the glass, the visible transmission would increase without affecting the other properties, thereby enhancing the FOM values for all the studied structures, as reflection losses on the glass side would be prevented. For example, in the case of 5.9 nm of Ag with a $Ag(O_2)$ seed layer, the T_v value would increase from 86.2 % to approximately 89.3 %, assuming the anti-reflective coating has a reflection of 0.5 %, thus resulting in a FOM of 25.0 $m\Omega^{-1}$ instead of 17.5 $m\Omega^{-1}$.

4. Conclusions

In this study, ultrathin silver samples deposited by sputtering on glass substrates using seed layers were analyzed. These seed layers were obtained by introducing small amounts of O_2 or N_2 during the initial silver growth, and the goal of the study was to find the optimal gas flow

Table 4
Roughness properties of the samples with and without O_2 additive in the seed layer or in the Ag layer itself.

Thickness (nm)	O_2 flow (sccm)	Roughness (nm)	Min. value (nm)	Max. value (nm)	Average roughness (nm)	Average height (nm)
4.5	0	1.33	0	12.9	1.05	6.2
4.5	30	0.53	0	5.3	0.41	1.7
4.5	20	0.69	0	6.9	0.53	2.0
4.5	50	0.80	0	7.3	0.61	2.4
2.5	0	0.97	0	8.8	0.77	3.7
2.5	50	0.44	0	8.0	0.31	1.7

Table 5
Roughness properties of samples with N_2 additive in the seed layer or in the Ag layer itself.

Thickness (nm)	N_2 flow (sccm)	Roughness (nm)	Min. value (nm)	Max. value (nm)	Average roughness (nm)	Average height (nm)
4.5	80	0.63	0	6.7	0.49	3.3
2.5	100	0.51	0	5.2	0.40	2.5

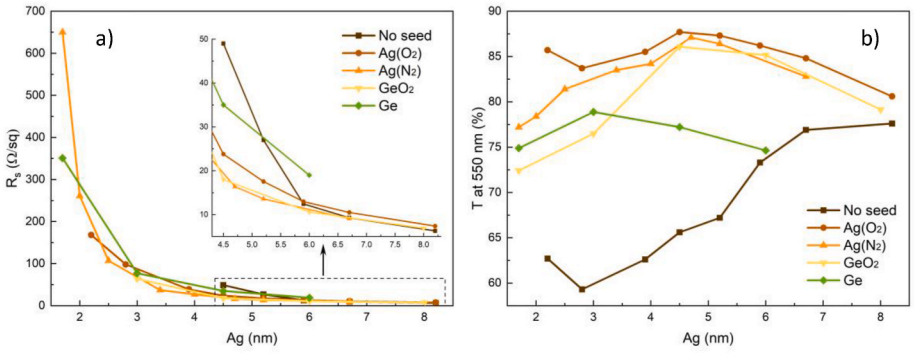


Fig. 12. (a) R_s and (b) T at 550 nm as a function of Ag thickness for pure Ag and for each seed layer deposited before Ag.

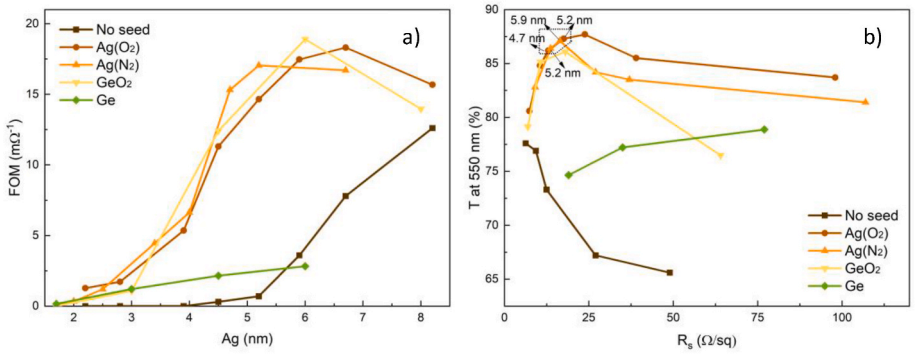


Fig. 13. (a) FOM as a function of Ag thickness and (b) T at 550 nm versus R_s for pure Ag and for each seed layer.

that would maximize the photoelectric properties of the entire structure. The gas flow was varied for different silver thicknesses, their transmission and resistance properties were measured, and it was concluded that a lower silver thickness required a higher gas flow. For example, for 4.5 nm of silver, the optimal O_2 flow was 30 sccm (partial pressure of 1.7×10^{-4} mbar), while for 3.9 nm, it was 40 sccm (partial pressure of 2.2×10^{-4} mbar). In the case of N_2 gas, for a silver thickness of 4.7 nm, the optimal N_2 flow was 80 sccm (partial pressure of 4.2×10^{-4} mbar), while for 4.0 nm, it was 100 sccm (partial pressure of 5.2×10^{-4} mbar). The surface morphological properties of the samples were also studied, and an improvement in surface roughness was observed when using seed layers at their optimal point compared to depositing pure silver. This improvement in the growth of the silver layer leads to enhanced properties of the studied samples and is responsible for the reduction in the threshold thickness of these ultrathin silver layers.

The results obtained with these two types of seed layers were compared with those obtained using Ge and GeO_2 seed layers. Superior results were obtained with the seed layers under study compared to the seed layers composed of metals or metal oxides (except for 6 nm of Ag). Comparing the results obtained with the $Ag(O_2)$ and $Ag(N_2)$ seed layers, it can be concluded that, in this deposition system, the optimal thicknesses leading to optimal T_v and R_s properties range between 5.9 and 4.7 nm of silver. For the $Ag(O_2)$ seed layer, the optimal thicknesses are 5.9 nm and 5.2 nm with 30 sccm of O_2, achieving 13 Ω/sq and 86.2 % and 17.6 Ω/sq and 87.3 % for R_s and T_v, respectively, for these two cases. When using $Ag(N_2)$ as the seed layer, the optimal thicknesses are 5.2 nm and 4.7 nm, with R_s and T_v values of 13.6 Ω/sq and 86.4 % and 16.4 Ω/sq and 87.1 %, respectively. When we examined the FOM values, the highest values were achieved with 6.7 nm of Ag using the $Ag(O_2)$ seed layer (18.3 $m\Omega^{-1}$), with 5.2 nm of Ag using the $Ag(N_2)$ seed layer (17.0 $m\Omega^{-1}$), and with 6 nm of Ag using the GeO_2 seed layer (18.9 $m\Omega^{-1}$).

CRediT authorship contribution statement

Ana Cueva: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Enrique Carretero:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Enrique Carretero reports financial support was provided by Spain Ministry of Science and Innovation. Enrique Carretero reports financial support was provided by European Union. Enrique Carretero reports financial support was provided by Government of Aragon Department of Science Technology and University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.vacuum.2024.113949.

Data availability

Data will be made available on request.

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